

ROBOTECH

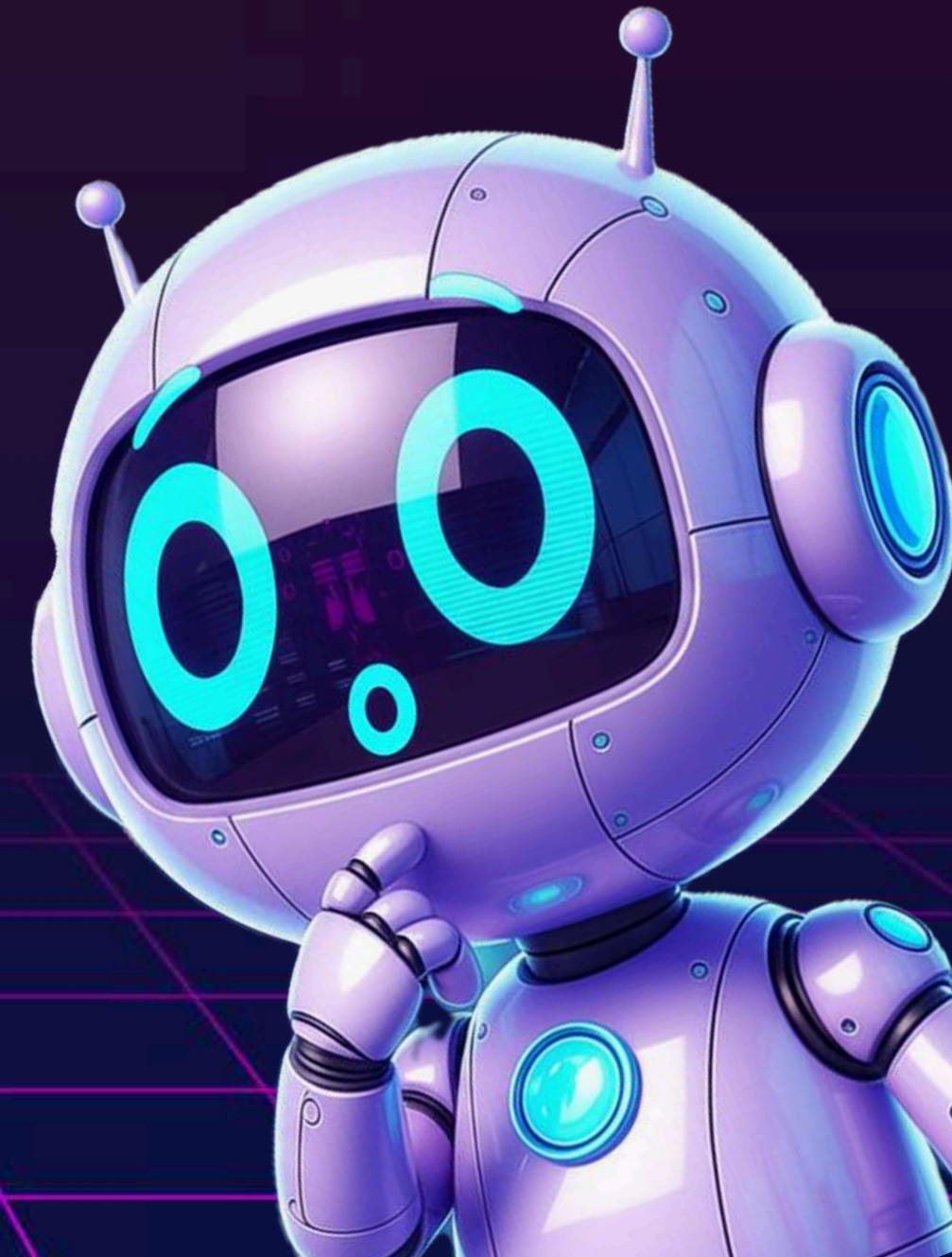
SESSION 0

INTRO TO PROGRAMMING



AGENDA

- » What is Programming
- » Machine Language
- » Programming Basics



What Is Programming



PROGRAMMING

giving instructions to a computer to make it do what you want — step by step.

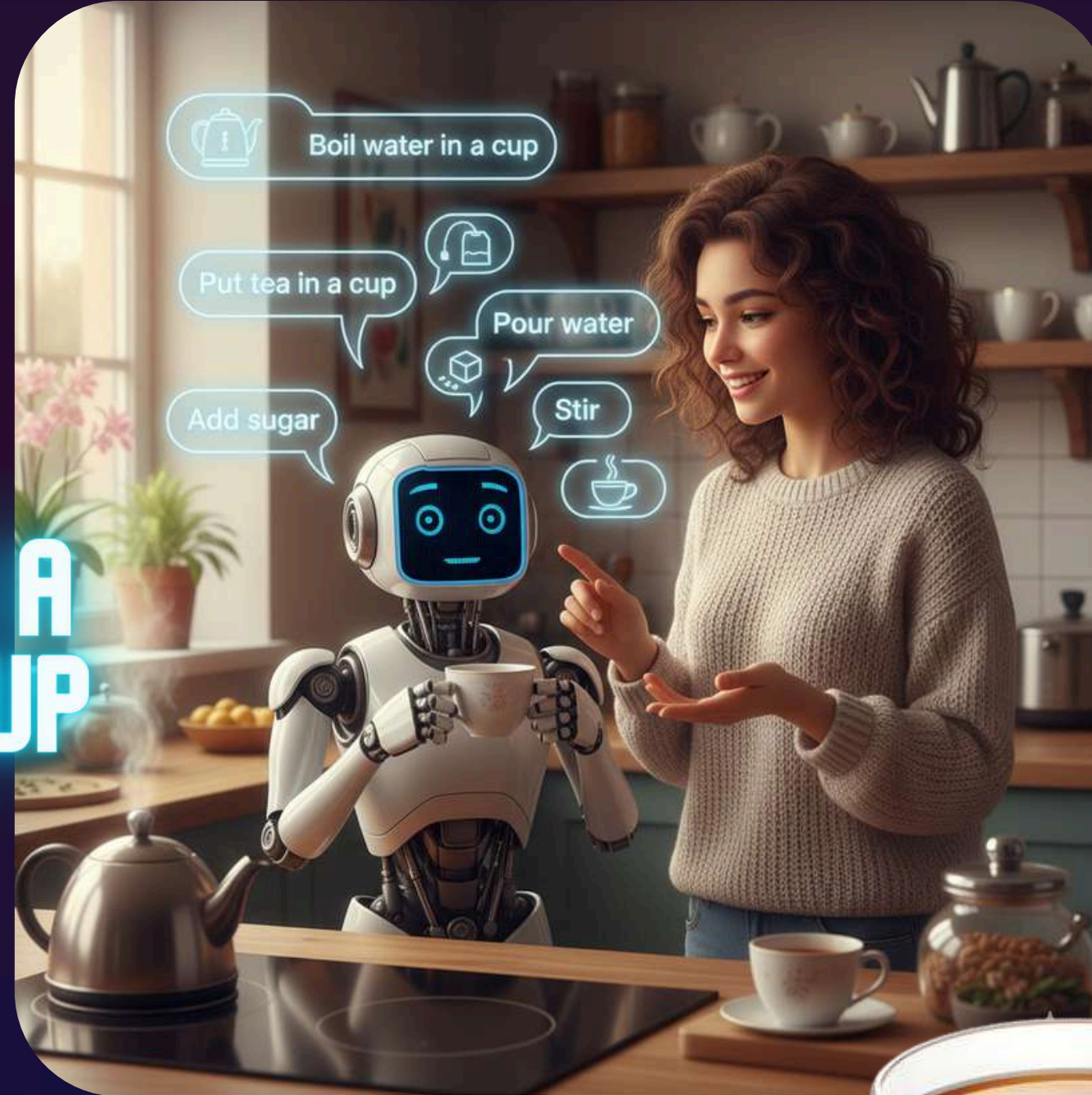
It's like teaching a computer how to think and what to do, but in a language it can understand.



IMAGINE YOU'RE TELLING A CHILD
HOW TO MAKE A CUP OF TEA.



IMAGINE YOU'RE TELLING A
CHILD HOW TO MAKE A CUP
OF TEA.



MAKE A CUP OF TEA.

You'd say:

Boil water

Put tea in a cup

Pour water

Add sugar

Stir

Those are clear, ordered steps — just like code!
Programming is writing those steps, but for a computer.





HOW COMPUTERS UNDERSTAND US

To give orders to someone, you need to speak their language.

🧠 Humans speak English, Arabic, etc.

💻 **But computers are machines — they don't understand words, they understand electricity ⚡**

- Inside the computer, everything is ON (1) or OFF (0) — just like a light switch! 💡
- So the computer's real "language" is made of 0s and 1s — we call it Binary Code.





EXAMPLE:

If we want to tell the computer “Hello” we could (in theory) write it like this:

Copy code 

```
01001000 01100101 01101100 01101100 01101111
```

That's just electric signals – not fun for humans

WE CAN'T TELL THE COMPUTER DIRECTLY WHAT TO DO.
WE NEED SOMETHING TO TRANSLATE OUR CODE INTO 0S AND 1S.



WHAT ARE PROGRAMMING LANGUAGES?

Programming languages are special languages that look more like human language but

- have strict grammar (called syntax)
- and special Vocab (reserved words) that have fixed meanings.

EXAMPLES: DART, PYTHON, JAVA, C++

But computers can't understand this language either.



THE TRANSLATOR : “ COMPILER ”

THE COMPILER IS THE TRANSLATOR THAT TAKES OUR CODE AND TURNS IT INTO MACHINE LANGUAGE (0S AND 1S).

```
print("Hello World!");
```

ComPiler

Copy code 

01001000 01100101 01101100 01101100 01101111

PC



FIRST LINE OF CODE



FIRST LINE OF CODE

Programming Statement :

A statement is a single instruction that tells the computer what to do.

Each statement is like one sentence in human language it must end properly and make sense.

This is a reserved vocab means "show an output"

START a normal language word

```
print("Hello World!");
```

Text to be Printed

end of Statement Sign

This is one statement



MAIN FUNCTION

```
2 void main() {  
3     // you can make a comment and write what you want in this line  
4     print("Hello World");  
5 }
```



Variables



VARIABLES

They serve as the storage boxes in our memory, capable of holding data and displaying it whenever we desire.

Fun fact: There are various sizes of boxes available.



DATATYPES

INT

INTEGER VALUES
NO LARGER THAN
32 BITS, VALUES
CAN BE FROM -2^{31}
TO 2^{31} .

DOUBLE

DECIMAL NUMBERS
ITS VALUES
DEPENDS ON THE
PLACE OF DECIMAL
POINT

STRING

USED IN STORING
WORDS OR
SENTENCES WITH
INFINITE LENGTH

BOOL

USED IN STORING
FACTS OR DATA
WITH TRUE OR
FALSE VALUES

OUR FAMOUS DATATYPES

```
3 // OUR KNOWN DATATYPES
4 String name    ="Shady";
5 int age =20;
6 double height = 180.52;
7 bool isMarried =false;
```



CONSOLE INPUT IN DART



INPUT IN DART

- To take user input, we use the `stdin.readLineSync()` function from the `dart:io` library.

```
dart.dart > ...  
1  import 'dart:io';  
2  
   Run | Debug  
3  void main() {  
4      print("Enter your name:");  
5      String? name = stdin.readLineSync();  
6      print("Hello, $name!");  
7  }
```

```
int age = int.parse(stdin.readLineSync()!);
```



Operators



ARITHMETIC OPERATORS

» + - *
/ %

COMPARISON OPERATORS

» == != > < >= <=

ASSIGNMENT

» = += -= *= /=




```
void main() {  
    int x = 5;  
    int y = 3;  
    print(x + y); // 8  
    print(x > y); // true  
    print(x == y); // false -> 5 not equal to 3  
    x+=2; // x = x + 2  
    print(x); // 7  
}
```



CONDITIONS (DECISION MAKING)



CONDITIONS

How can we make our program decide what to do?

Giving the computer a choice

If this happens → do that, else → do something else.



```
void main() {  
    int age = 22;  
    if (age >= 18) {  
        print("You can vote!");  
    } else {  
        print("Too young to vote.");  
    }  
}
```



LOGICAL OPERATORS

» $\&\& \rightarrow$ AND

$\|\rightarrow$ OR

$!\rightarrow$ NOT



```
void main() {  
    bool isSunny = true;  
    bool hasSunGlasses = false;  
  
    if (isSunny && ! hasSunGlasses) {  
        print("Don't Go outside!");  
    }  
}
```



Loops



LOOPS



“What if we want to repeat an action many times?”

```
void main() {  
    print("Hello, World!");  
    print("Hello, World!");  
    print("Hello, World!");  
    print("Hello, World!");  
    print("Hello, World!");  
}
```



```
void main() {  
    for (var i = 0; i < 5; i++) {  
        print("Hello, World!");  
    }  
}
```



LOOPS TYPES

FOR

USED WHEN I
KNOW THE
NUMBER OF
REPEATS

WHILE

USED WHEN I WANT
TO REPEAT UNTIL
SPECIFIC
CONDITION

DO-WHILE

SAME AS WHILE
BUT I MUST DO
INSTRUCTIONS AT
LEAST ONCE

LOOPS

FOR LOOP

```
2 void main() {  
3  
4     for(int i=1; i<=5;i++){  
5         print("Hello World");  
6     }  
7  
8 }
```



LOOPS

WHILE

```
void main() {  
  
    while(true){  
        print("infinite Loop");  
    }  
}
```



LOOPS

DO WHILE

```
1  
2 void main() {  
3     do{  
4         print("just one loop");  
5     }  
6     while(false);  
7 }
```



COLLECTIONS



COLLECTIONS

“How do we store and use multiple values together?”

→ LIST

→ SET

→ MAP



LISTS EXAMPLES

```
void main() {  
    String s1 = "Ahmed";  
    String s2 = "Shady";  
    String s3 = "Adham";  
    String s4 = "Mostafa";  
  
    List students = [s1 , s2 , s3 , s4];  
  
    print(students);  
  
    print(students[1]);  
}
```



Functions



LOOPS

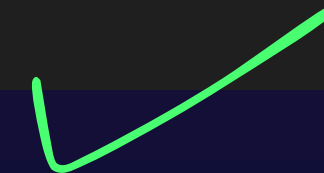


“What if we want to repeat an action many times?”

```
void main() {  
    int a = 1;  
    int b = 2;  
    print(a + b);  
  
    a = 5;  
    b = 3;  
    print(a + b);  
  
    a = 4;  
    b = 9;  
    print(a + b);  
}
```



```
void main() {  
    printSum(1 , 2);  
    printSum(5 , 3);  
    printSum(4 , 9);  
}  
  
void printSum(int a , int b){  
    print(a + b);  
}
```



ROBOTECH

THANK YOU



علمٌ ينتفعُ به